

DEVELOPMENT ARCHITECTURE

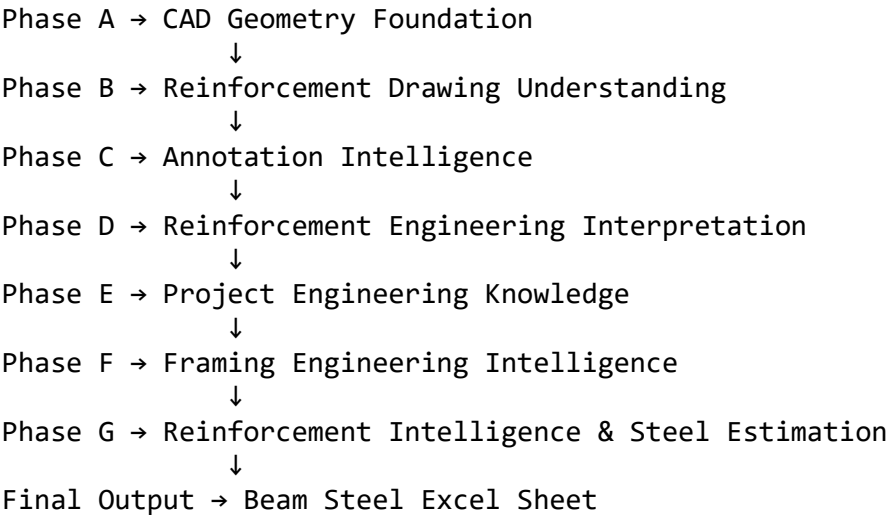
CODE MAP

Current Production Architecture — Version10 Hybrid W.19.1

AUTHORITY • PRODUCTION_TRUTH.md (Version10 Hybrid W.19.1)
SCOPE • Important live production files under Version10/.
STATUS • Historical Version1–Version9 trees are not current production.

1. Architecture at a Glance

The project has evolved through the following major phases:



A quick-read map before the detailed file inventory.

Current production flow

DXF upload
VROOT1 → R1 + T1 + R.3.1 → R2A + R21B–D → L.2.2 + R3 + R.1.2A
→ R.1.3 + pieces + M.2 → W.6 Hybrid → VB.1 → Steel + BBS + Excel

RULE OF AUTHORITY
Vision decides what bars exist (count / diameter / role).
Deterministic engineering (R.1.3 / VB.1) decides how they are cut, spaced, hooked, weighed, and written to Excel.

Important interpretation

- Phases A–G are the original development architecture. They are capabilities, not the current web stage IDs.
- The live estimator runs 14 PRODUCTION_STAGES that implement those capabilities.
- Paths in this document are relative to Version10/ unless noted.
- Orchestration (not a historical A–G phase): webapp/ + Run_PY/ + src/config/run_context.py.

Historical A–G → Live 14-stage mapping

Historical phase	Live stage(s)	Package / primary area
A — CAD Geometry	VROOT1, L.2.2, R3,	PhaseVROOT.1_*, PhaseL.2.2_*, PhaseR3_*,

Foundation	R.1.2A	PhaseR1_2A_*
B — Reinforcement Drawing Understanding	R1, T1	PhaseR.1_*, PhaseT1_*
C — Annotation Intelligence	R1, R.3.1	PhaseR.1_*, PhaseR3.1_*
D — Reinforcement Engineering Interpretation	R21B, R21C, R21D	PhaseR2.1B_*, PhaseR2.1C_*, PhaseR2.1D_*
E — Project Engineering Knowledge	R2A	PhaseR.2A_*
F — Framing Engineering Intelligence	L.2.2, R3, R.1.2A	Same as A geometry chain
G — Reinforcement Intelligence & Steel Estimation	R.1.3, W.6, VB.1	PhaseR1.3_*, PhaseW6_*, PhaseVB.1_*
Final Excel	VB.1	estimator_excel_generator.py

2. Phase A — CAD Geometry Foundation

Intent: Read DXFs, classify drawings, discover beams, and recover span/section geometry.

LIVE STAGES • VROOT1 → L.2.2 → R3 → R.1.2A

VROOT1 — Project / drawing / beam discovery

Code File	Description
Run_PY/run_phase_vroot1_dynamic_pipeline_initialization.py	Web stage 1 runner.
src/PhaseVROOT.1_dynamic_pipeline_initialization/phase_vroot1_orchestrator.py	Sequences discovery registry and 8 canonical JSON artefacts.
project_discovery.py	Finds framing, reinforcement, and general-notes DXFs in the upload folder.
drawing_classifier.py	Labels each DXF as framing / reinforcement / GN.
drawing_manifest_builder.py	Writes the drawing inventory for the run.
dynamic_beam_discovery.py	Reads framing DXF geometry/text to find beam IDs.
beam_registry_builder.py	Builds beam_registry.json (canonical beam list).
engineering_object_initializer.py	Creates empty engineering object shells for every discovered beam.
initialization_export.py	Writes run-scoped VROOT1 JSON under STEEL_OUTPUT_ROOT.

L.2.2 / R3 / R.1.2A — Geometry recovery

See Phase F — Framing Engineering Intelligence for the live geometry-recovery file map.

3. Phase B — Reinforcement Drawing Understanding

Intent: Find beam-detail regions, sketches, text, and stirrup geometry on the reinforcement DXF.

LIVE STAGES • R1 → T1

File	Description
Run_PY/run_phase_r1_generalized_reinforcement_discovery.py	Web stage 2 runner.
src/PhaseR.1_generalized_reinforcement_discovery/phase_r1_orchestrator.py	Full R.1 pass: details, annotations, groups.
beam_detail_discovery.py	Locates beam-detail regions on the reinforcement drawing.
beam_detail_segmenter.py	Splits continuous detail strips into per-beam segments.
annotation_discovery.py	Collects DXF TEXT/MTEXT that look like bar marks.
dxf_text_utils.py	Normalizes CAD text (Y12, 2-Y16, stirrup marks).
reinforcement_geometry_mapper.py	Maps annotations onto nearby CAD geometry.
Run_PY/run_phase_t1_geometric_stirrup_evidence.py	Web stage 3 runner.
src/PhaseT1_geometric_stirrup_evidence/phase_t1_orchestrator.py	Builds geometric stirrup evidence and crop envelopes for Vision.
vector_stirrup_detector.py	Detects stirrup symbols from DXF vectors (not Vision).
geometry_envelope.py	Beam crop envelopes later used by W.8 evidence PNGs.
beam_extent.py	Estimates beam length extent from CAD.
type3_label_repair.py	Repairs truncated Type-3 stirrup labels (@100; @100/200/100).

4. Phase C — Annotation Intelligence

Intent: Group annotations, own them to a beam, attach leaders/arrows, and prepare engineering meaning.

LIVE STAGES • R1 (grouping / ownership) + R.3.1 (leaders / relationships)

File	Description
reinforcement_annotation_classifier.py	Classifies a text mark (main bar, extra, stirrup, SFR, spacer).
reinforcement_group_builder.py	Groups related marks into one reinforcement object.
adaptive_association_engine.py	Associates nearby text, geometry, and beam IDs when layout is messy.
reinforcement_relationship_builder.py	Links annotations to sketches / leaders.
reinforcement_role_classifier.py	Assigns TOP/BOTTOM/SIDE and MAIN/EXTRA

	roles.
engineering_reinforcement_builder.py	Emits the R.1 engineering reinforcement model JSON.
Run_PY/run_phase_r31_engineering_relationship_engine.py	Web stage 10 runner.
src/PhaseR3.1_engineering_relationship_engine/phase_r31_orchestrator.py	Builds drawing-level annotation geometry relationships.
leader_discovery.py	Finds leader lines from annotations.
leader_chain_builder.py	Chains leader segments to the pointed bar.
arrow_detector.py	Detects arrowheads that indicate ownership.
physical_bar_detector.py	Finds drawn bar lines in the detail.
annotation_relationship_builder.py	Owns each annotation to a bar / beam.
relationship_graph_builder.py	Graph of beam–annotation–geometry links.

6. Phase D — Reinforcement Engineering Interpretation

Intent: Turn classified marks into engineering meaning: role, quantity, placement, and conflicts.

LIVE STAGES • R21B → R21C → R21D

File	Description
Run_PY/run_phase_r21b_semantic_interpreter.py	Web stage 5; loads Version10 R.2.1B (not Version8).
src/PhaseR2.1B_engineering_semantic_interpreter/phase_r21b_orchestrator.py	Interprets R.1 marks using the semantic dictionary.
semantic_interpreter.py	Core “what does this mark mean?” engine.
semantic_role_resolver.py	Resolves bar role (top main, extra, stirrup, ...).
semantic_quantity_resolver.py	Resolves count / spacing from the mark.
semantic_placement_resolver.py	Resolves where on the beam the bar sits.
semantic_conflict_resolver.py	Chooses among competing readings of the same mark.
engineering_meaning_builder.py	Writes engineering_semantic_objects.json.
Run_PY/run_phase_r21c_engineering_fact_normalization.py	Web stage 6.
src/PhaseR2.1C_engineering_fact_normalization/phase_r21c_orchestrator.py	Normalizes semantics into typed engineering facts.
role_normalizer.py / placement_normalizer.py / intent_normalizer.py	Canonicalise role, placement, and intent fields.
engineering_fact_builder.py	Builds EngineeringFacts.json.
Run_PY/run_phase_r21d_evidence_hypothesis_engine.py	Web stage 7.
src/PhaseR2.1D_evidence_hypothesis_engine/phase_r21d_orchestrator.py	Scores competing hypotheses with evidence.
evidence_builder.py	Attaches CAD / annotation evidence to each fact.
hypothesis_ranker.py	Ranks hypotheses; later T.1 / Vision can confirm

stirrups.

Historical D.4 / D.4.1 / D.4.2 annotation parsers live in older trees. Live interpretation is R.2.1B–D plus R.1.3.

7. Phase E — Project Engineering Knowledge

Intent: Read General Notes for Ld tables, cover, grades, hooks, and laps — project defaults with provenance.

LIVE STAGE • R2A

File	Description
Run_PY/run_phase_r2a_engineering_context.py	Web stage 4; Version10 GN factory.
src/PhaseR.2A_engineering_context/phase_r2a_orchestrator.py	Parses GN DXF into Engineering Context.
engineering_context_factory.py	GN search order: uploaded STEEL_RUN_ROOT/general_notes registry pointer → Version10 Benchmark_Set_2.
general_notes_text_extractor.py	Pulls all text from the GN drawing (including blocks).
general_notes_classifier.py	Splits GN text into Ld / cover / grade / hook / lap topics.
development_length_parser.py	Parses development-length tables (Ld).
cover_parser.py	Parses clear cover.
steel_grade_parser.py / concrete_grade_parser.py	Parses Fe415/Fe550 and concrete grade.
hook_rule_parser.py / lap_rule_parser.py	Parses hook multiples and lap lengths.
engineering_context_builder.py	Assembles the typed Engineering Context object.
engineering_context_writer.py	Writes run-scoped R.2A JSON used later by R.1.3 / VB.1.

8. Phase F — Framing Engineering Intelligence

Intent: Recover supports, spans, axes, engineering lengths, and a validated beam section.

LIVE STAGES • L.2.2 → R3 → R.1.2A

File	Description
Run_PY/run_phase_l2_2_geometry_recovery.py	Web stage 8 (L.2.2, not historical L.2).
src/PhaseL.2.2_geometry_recovery/phase_l22_orchestrator.py	Recovers geometry registry from VROOT1 beam_registry.json.
geometry_registry.py	Per-beam span / depth / width / support slots.
geometry_registry_engine.py	Fills those slots from framing CAD.
Run_PY/run_phase_r3_geometry_context_engine.py	Web stage 9.
src/PhaseR3_geometry_context_engine/phase_r3_orchestrator.py	Builds GeometryContexts.json (axis, zones, supports).

beam_axis_builder.py	Beam centreline / axis.
support_locator.py / support_zone_classifier.py	Support locations and support vs span zones.
span_zone_classifier.py	Mid-span vs support shear zones (needed for stirrups).
normalized_position_builder.py	Station 0–1 along the beam.
extent_evidence_builder.py	Evidence for how far a bar extends.
Run_PY/run_phase_r12a_geometry_accuracy.py	Web stage 11.
src/PhaseR1_2A_geometry_accuracy/phase_r12a_orchestrator.py	Validates / freezes validated_beam_geometry.json.
geometry_provider.py	Single geometry source for downstream R.1.3.
geometry_validators.py	Span / section sanity checks.

9. Phase G — Reinforcement Intelligence & Steel Estimation

Intent: Own every bar, compute pieces / cut lengths / spacers, overlay Vision semantics when needed, then produce steel kg + BBS + Excel.

G.5 • R.1.3 engineering model | HYBRID • W.5 / W.8 / Vision / D.2 / W.6 | G.6–G.7 • VB.1 output

G.5 — R.1.3 Engineering Model (Live Stage 12)

File	Description
Run_PY/run_phase_r13_pipeline_integration.py	Web stage 12; loads Version10 R.1.3 and nested packages.
src/PhaseR1.3_pipeline_integration/phase_r13_orchestrator.py	Integrates facts + geometry into production bar models.
pipeline_integration_manager.py	Dynamically loads pieces, R.1.2B, R.1.2C/D, and M.2 spacers.
engineering_bar_builder.py	Builds EngineeringBarModel per bar from pieces / details.
engineering_bar_model.py	Production bar record: role, dia, qty, Ld, hooks, ...
reinforcement_source_selector.py	Chooses which upstream artefact is the reinforcement source.
src/PhaseR1_3_reinforcement_piece_generation/piece_builder.py	Turns details into fabricable pieces (loaded inside R.1.3, not a web stage).
piece_geometry.py	Piece shape: straight, crank, hook.
piece_quantity.py	Piece counts.
src/PhaseV9_spacer_rule/spacer_engine.py	M.2 spacer / cover bars from geometry + cover.
r13_injector.py	Injects spacer bars into the R.1.3 production model.
src/PhaseSl.1_stirrup_improvement/phase_si1_orchestrator.py	StirrupImprover.compute_beam(): zone-wise stirrup BBS rows used by VB.1 (Version10 copy).

Hybrid Overlay — W.5 / W.8 / Vision / D.2 / W.6 (Live Stage 13)

This is the current semantic path. It does NOT recompute cut length, stirrup quantity, kg, or Excel.

File	Description
Run_PY/run_phase_w6_hybrid_production_authority.py	Web stage 13.
src/PhaseW8_production_vision_evidence/generator.py	Builds context + detail PNG evidence from T.1 envelopes.
src/PhaseW5_production_hybrid_shadow/adapter.py	Wires evidence → Claude D.2 resolver.
live_invoke.py	Places the live Claude Vision call (fail-closed).
semantic.py	Applies Vision vs deterministic field authority.
src/PhaseP2610C5_*/vision_prompt.py	Production Vision prompt.
claude_call.py	Constrained JSON Vision call.
src/PhaseP253_*/claude_vision_client.py	Anthropic Vision client wrapper.
src/PhaseP2610E2_*/live_caller.py	Retry / fail-closed live caller used by W.5.
src/PhaseP2610C3_*/claude_client.py	encode_png for image parts.
src/PhaseM.1_engineering_vision_dataset/dxf_renderer.py	Renders DXF regions to PNG.
src/PhaseP2610D2_*/resolver.py	Hybrid semantic resolution (D.2).
src/PhaseW6_hybrid_production_authority/orchestrator.py	run_production_hybrid — production Hybrid entry.
handoff.py	Patches count / diameter / role onto R.1.3 JSON; protects cut length, stirrup qty, geometry, spacers, kg, BBS.
visuals.py	Hybrid coverage / visual completeness helpers.

G.6 / G.7 — Steel, BBS, Excel (Live Stage 14)

File	Description
Run_PY/run_phase_vb1_production_output_completion.py	Web stage 14.
src/PhaseVB.1_production_output_completion/phase_vb1_orchestrator.py	Steel → BBS → Excel; reads run-scoped R.1.3 (+ Hybrid patches).
steel_weight_completion.py	Cut length × unit weight kg (uses GN Ld / cover when present).
bbs_completion_engine.py	Estimator-style BBS rows (stirrups via Sl.1 StirrupImprover).
estimator_excel_generator.py	Writes Production_Output/Estimation_Output.xlsx.
excel_structure_builder.py	Workbook sheets / layout.
worksheet_formatter.py	Estimator formatting.
workbook_validator.py	Checks the workbook is a valid deliverable.

FINAL ARTEFACT

<run>/data/output/Production_Output/Estimation_Output.xlsx

10. End-to-End Current Hybrid Pipeline

The sequence below is the clearest operational view of the live estimator.

- 1 DXF upload
- 2 VROOT1 – discovery (A, G.1)
- 3 R1 – annotation / groups (B, C)
- 4 T1 – stirrup geometry + envelopes (B)
- 5 R2A – General Notes context (E)
- 6 R21B–D – semantic facts (D)
- 7 L.2.2 / R3 / R.1.2A – framing geometry (A, F)
- 8 R.3.1 – leaders / ownership (C)
- 9 R.1.3 – bars + pieces + spacers (G.5)
- 10 W.8 evidence → Claude D.2 → W.6 handoff (Hybrid overlay)
- 11 VB.1 – steel / BBS / Excel (G.6, G.7)

11. Explicitly Not Current Production

Do not treat the following as the live A–G implementation.

- Version1/ ... Version9/ and Steel-Beam-Estimation/
- Run_PY/run_phase_l2_engineering_reinforcement_interpretation.py — historical L.2; web uses L.2.2.
- Run_PY/run_phase_r13_reinforcement_piece_generation.py — not a web stage; pieces load inside R.1.3.
- Benchmark / QA CLIs: PhaseVA.2, PhaseVTEST*, PhaseVRUN.1, PhaseQA*.
- Remaining files inside Vision packages that are only benchmark orchestrators (see V10_Organize/PRODUCTION_MODULE_INDEX.md).

12. Quick Reference — Live Stage Sequence

Web stage	Live ID	Primary responsibility	Historical capability
1	VROOT1	Project / drawing / beam discovery	A / G.1
2	R1	Reinforcement discovery + annotation grouping	B / C
3	T1	Geometric stirrup evidence + envelopes	B
4	R2A	General Notes engineering context	E
5	R21B	Semantic interpretation	D
6	R21C	Engineering fact normalization	D
7	R21D	Evidence / hypothesis ranking	D
8	L.2.2	Geometry recovery	A / F
9	R3	Geometry context / supports / zones	A / F
10	R.3.1	Leaders / annotation relationships	C
11	R.1.2A	Geometry accuracy / validation	A / F
12	R.1.3	Production bar model + pieces + spacers	G.5
13	W.6 Hybrid	Vision semantic overlay / authority handoff	Hybrid
14	VB.1	Steel weight + BBS + Excel	G.6 / G.7